ISIC2018 segmentation with SAM

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1 Introduction

Skin lesion segmentation is an important task in the analysis of dermoscopic images. The objective is to identify the pixels corresponding to a skin lesion and separate them from the surrounding healthy skin. Accurate segmentation can provide useful information for subsequent computer-aided analysis, including lesion characterization and classification.

Despite its apparently simple formulation, skin lesion segmentation is challenging because dermoscopic images exhibit substantial variability. Lesions can differ considerably in size, shape, color, and texture. In addition, images may contain artifacts such as hair, illumination variations, reflections, and low-contrast lesion boundaries. These factors make it difficult for simple image-processing techniques to produce reliable segmentation masks across different images.

In this work, we investigate several approaches to skin lesion segmentation using the *ISIC 2018 Task 1* dataset. The experiments are designed to progressively investigate the effect of classical image processing, foundation-model-based segmentation, domain-specific fine-tuning, human interaction, and a dedicated semantic segmentation architecture.

The first experiment establishes a classical baseline based on Otsu thresholding combined with morphological post-processing. This provides a simple reference against which the learning-based approaches can be compared.

The second experiment investigates the zero-shot transfer capability of the Segment Anything Model (SAM). Two different prompting strategies are considered. First, SAM is provided with a bounding box derived from the ground-truth segmentation mask. This experiment is used to study the segmentation capability of SAM when accurate localization information is available. Second, an automatically generated bounding box is obtained from the classical Otsu segmentation and provided to SAM. This represents a more realistic automatic pipeline in which no ground-truth information is available during inference.

The third experiment investigates domain adaptation by fine-tuning only the SAM mask decoder on the ISIC training data while keeping the image encoder and prompt encoder frozen. Two corresponding evaluation settings are considered: fine-tuning and evaluation with ground-truth bounding boxes, and evaluation using automatically generated Otsu-derived bounding boxes.

The fourth experiment explores interactive segmentation. Instead of requiring the user to manually segment the entire lesion, the proposed procedure uses the Otsu segmentation as an initial estimate and asks for a minimal human correction only when the automatic segmentation differs sufficiently from the ground-truth mask during experimental evaluation. The correction consists of one positive point indicating lesion tissue and one negative point indicating background tissue. These points are then supplied to the fine-tuned SAM model as additional prompts.

Finally, the fifth experiment evaluates a dedicated end-to-end segmentation architecture based on U-Net with an ImageNet-pretrained ResNet-34 encoder. Unlike the SAM-based approaches, the U-Net directly predicts a segmentation mask from the input image without requiring a bounding-box or point prompt at inference time.

The experiments therefore investigate the following progression:

1. A classical, non-learning-based segmentation baseline.
2. Zero-shot segmentation using a general-purpose foundation model.
3. Domain-specific fine-tuning of the SAM mask decoder.
4. Interactive segmentation using minimal human correction.
5. Fully automatic segmentation using a dedicated U-Net architecture.

The objective is not only to compare segmentation performance, but also to understand the role of prompting, domain-specific fine-tuning, and human interaction in adapting a general-purpose segmentation model to dermoscopic images.

2 Dataset and Experimental Setup

2.1 ISIC 2018 Task 1 Dataset

All experiments in this work are conducted on the *ISIC 2018 Task 1* skin lesion segmentation dataset. The task consists of segmenting the lesion region in dermoscopic images by predicting a binary mask in which lesion pixels are assigned to the foreground and all other pixels belong to the background.

The dataset contains dermoscopic images together with manually annotated binary segmentation masks. Each image is associated with a corresponding ground-truth mask defining the lesion boundary.

For the experiments presented in this work, the locally available dataset was organized into training, validation, and test subsets. The resulting number of usable image–mask pairs is shown in Table 1.

Table 1: Dataset split used throughout the experiments.

Subset	**Number of images**
Training	2,594
Validation	100
Test	1,000
Total	**3,694**

The training subset is used for model optimization, while the validation subset is used for model selection and monitoring during training. The test subset is kept separate from training and model selection and is used only for the final evaluation of the trained methods.

In particular, the test images are not used to optimize model parameters, select training epochs, or determine hyperparameters. This separation is maintained across the different experiments in order to provide a consistent basis for comparison.

For the SAM fine-tuning experiments, the training images were loaded using a data loader rather than loading the entire dataset into memory simultaneously. This allows the experiments to scale to the complete training set while maintaining reasonable memory consumption.

2.2 Image and Mask Preprocessing

The original dermoscopic images have different spatial resolutions. To simplify the computational requirements and ensure a consistent input size across the different methods, all images are resized to 256×256 pixels.

The same spatial transformation is applied to the corresponding segmentation masks. Since the masks contain discrete class labels, nearest-neighbor interpolation is used when resizing the masks in order to avoid introducing intermediate label values.

After resizing, the ground-truth masks are converted into binary masks:

$$M(x, y) = \begin{cases} 1, & \text{if the pixel belongs to the lesion,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

For the classical segmentation baseline, the RGB images are converted to grayscale before applying the thresholding procedure. For the learning-based models, the RGB images are provided as input after the corresponding preprocessing required by each architecture.

2.3 Training and Validation Protocol

The training subset contains 2,594 image–mask pairs and is used to train the learning-based models. Because loading all training images simultaneously into memory is unnecessary, the training data are accessed using a data-loading pipeline. Images and masks are loaded as required during training rather than storing the complete training dataset in RAM.

The 100-image validation subset is used to monitor model performance during training. In particular, the validation Intersection over Union (IoU) is used as the primary criterion for selecting the best model checkpoint in the fine-tuning and U-Net experiments.

The 1,000-image test subset is reserved for the final evaluation. No test-set performance is used to select a model checkpoint.

This protocol can be summarized as follows:

Table 2: Role of each dataset subset in the experimental protocol.

Subset	**Purpose**
Training	Model parameter optimization
Validation	Model selection and checkpoint selection
Test	Final performance evaluation only

2.4 Evaluation Metrics

Two complementary metrics are used to evaluate segmentation performance: Intersection over Union (IoU) and Dice similarity coefficient (Dice).

2.4.1 Intersection over Union

Intersection over Union measures the overlap between a predicted segmentation mask and the corresponding ground-truth mask. It is defined as

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}, \tag{2}$$

where P denotes the predicted foreground mask and G denotes the ground-truth foreground mask.

The IoU value ranges from 0 to 1, where a value of 1 indicates perfect overlap between the prediction and the ground truth.

For each image, IoU is computed independently. The reported mean IoU is then obtained by averaging the per-image IoU values over the test set.

2.4.2 Dice Similarity Coefficient

The Dice coefficient is another commonly used measure of segmentation overlap. It is defined as

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}. \tag{3}$$

Similar to IoU, Dice ranges from 0 to 1, with larger values indicating greater agreement between the predicted and ground-truth masks.

Both metrics are reported because they provide complementary views of segmentation quality. IoU directly measures the ratio between intersection and union, whereas Dice places greater emphasis on the overlap between the predicted and ground-truth foreground regions.

For numerical stability, a small ϵ is added to the numerator and denominator in the implementation:

$$\text{IoU} = \frac{|P \cap G| + \epsilon}{|P \cup G| + \epsilon}, \tag{4}$$

and

$$\text{Dice} = \frac{2|P \cap G| + \epsilon}{|P| + |G| + \epsilon}. \tag{5}$$

In addition to the mean values, the standard deviation, median, minimum, and maximum values are reported where appropriate in order to describe the variability of segmentation performance across individual test images.

2.5 Experimental Categories

The experiments are organized into three broad categories according to the information available to the segmentation model at inference time.

First, **automatic segmentation** refers to methods that require no manually supplied spatial prompt during inference. The classical Otsu baseline and the U-Net belong to this category. The automatic SAM pipeline also belongs to this category because its bounding box is generated automatically from the Otsu segmentation.

Second, **oracle-prompted segmentation** refers to experiments in which information derived from the ground-truth mask is supplied to the model as a prompt. The SAM experiments using a ground-truth bounding box fall into this category. These experiments are not intended to represent a deployable automatic pipeline. Instead, they provide an estimate of the segmentation capability of SAM when accurate localization information is available.

Third, **interactive segmentation** refers to the experiment in which a human provides a small amount of corrective information through positive and negative points. This setting is investigated in Step 4.

This distinction is important because the amount of information available to the model differs between these settings. Consequently, the corresponding results should be interpreted as different experimental conditions rather than as a simple ranking of models.

2.6 Overall Experimental Pipeline

The complete experimental study consists of five main steps.

1. **Step 1 – Classical baseline:** Otsu thresholding followed by morphological processing and connected-component selection is used to obtain a traditional image-processing baseline.
2. **Step 2 – Zero-shot SAM:** The pretrained Segment Anything Model (SAM) is evaluated without domain-specific training. Two prompt types are considered:
   - a ground-truth bounding box;
   - an automatically generated bounding box obtained from the Otsu segmentation.
3. **Step 3 – SAM fine-tuning:** The SAM mask decoder is fine-tuned using the ISIC training data while the image encoder and prompt encoder remain frozen. Both prompting conditions are considered:
   - ground-truth bounding boxes;
   - automatically generated Otsu-derived bounding boxes.
4. **Step 4 – Interactive SAM:** The automatically generated Otsu segmentation is used as the starting point. When the automatic segmentation differs sufficiently from the ground truth during the experimental evaluation, a minimal human correction consisting of one positive and one negative point is introduced. These points are supplied to the fine-tuned SAM model.
5. **Step 5 – U-Net:** A dedicated U-Net segmentation architecture with an ImageNet-pretrained ResNet-34 encoder is trained on the ISIC training data. Unlike the SAM-based approaches, the U-Net directly predicts a segmentation mask without requiring a spatial prompt.

This progressive design makes it possible to distinguish the effects of three different factors: classical segmentation, the use of a foundation model, and task-specific training.

2.7 Implementation Environment

The experiments were implemented in Python using common computer vision and deep learning libraries. OpenCV was used for image processing and the classical Otsu segmentation pipeline. PyTorch was used for model training and inference. The Segment Anything implementation was used for the SAM experiments, while the U-Net experiment was implemented using the `segmentation_models_pytorch` framework.

GPU acceleration was used when available. The SAM experiments used the ViT-B variant of the Segment Anything Model. The U-Net experiment used a ResNet-34 encoder initialized with ImageNet-pretrained weights.

The same test subset and evaluation metrics were used throughout the experiments to facilitate comparison between the different approaches.

3 Step 1: Classical Otsu Segmentation Baseline

3.1 Objective

The first experiment establishes a classical image-processing baseline for skin lesion segmentation. The purpose of this experiment is to determine how well lesion segmentation can be performed without machine learning or a foundation model.

The baseline is based on intensity thresholding using Otsu's method, combined with several preprocessing and morphological operations. The resulting segmentation provides a reference point for evaluating the subsequent SAM-based and U-Net-based approaches.

The complete processing pipeline is:

RGB image
↓
grayscale conversion
↓
hair removal
↓
Otsu thresholding
↓
morphological processing
↓
largest connected component
↓
binary lesion mask

3.2 Grayscale Conversion

The input RGB image is first converted to a grayscale image. The motivation is to obtain a single intensity channel on which the thresholding operation can be performed.

Let I_R, I_G, and I_B denote the red, green, and blue channels. The grayscale image is obtained using the standard RGB-to-grayscale transformation implemented by OpenCV.

The resulting image is denoted by

$$I_{\text{gray}}(x, y). \tag{6}$$

Because many skin lesions are visually darker than their surrounding skin, intensity-based thresholding provides a simple first approximation for separating the lesion from the background.

3.3 Hair Detection and Removal

Dermoscopic images frequently contain dark hair structures crossing the skin and lesion regions. Since the classical segmentation method relies on dark pixels to identify the lesion, hair can be incorrectly classified as lesion foreground.

To reduce this effect, a black-hat morphological transformation is applied to the grayscale image. A black-hat transformation emphasizes dark structures that are smaller than the chosen structuring element.

Let K denote an elliptical structuring element. The black-hat image is defined as

$$I_{\mathrm{BH}} = (I_{\mathrm{gray}} \bullet K) - I_{\mathrm{gray}}, \tag{7}$$

where $\bullet$ denotes the morphological closing operation.

In the implementation, an elliptical 9×9 structuring element is used. The black-hat image is subsequently thresholded to obtain a binary estimate of dark hair structures.

The detected hair pixels are then removed using image inpainting. The Telea inpainting algorithm is used to reconstruct the affected regions from their surrounding pixels.

The resulting image is denoted as I_{clean}.

This preprocessing step is intended to reduce the influence of hair artifacts on the subsequent thresholding operation.

3.4 Otsu Thresholding

After preprocessing, Otsu's thresholding method is applied to the cleaned grayscale image.

Otsu's method automatically determines an intensity threshold that separates the image into two classes by minimizing the within-class variance, or equivalently maximizing the between-class variance.

Because the lesion is assumed to be darker than the surrounding skin, the inverse binary thresholding mode is used. The resulting binary image is

$$B(x, y) = \begin{cases} 1, & I_{\mathrm{clean}}(x, y) < T_{\mathrm{Otsu}}, \\ 0, & \text{otherwise}, \end{cases} \tag{8}$$

where T_{Otsu} is the threshold automatically determined by Otsu's method.

This operation produces an initial estimate of the lesion region.

3.5 Morphological Post-processing

The raw thresholded mask may contain small isolated regions, holes, and irregular boundaries. Morphological operations are therefore applied to improve the spatial coherence of the predicted mask.

An elliptical 7×7 structuring element is used.

First, morphological opening is applied. Opening consists of an erosion followed by a dilation and is primarily used to remove small isolated foreground structures.

Second, morphological closing is applied. Closing consists of a dilation followed by an erosion and is used to fill small gaps and connect nearby foreground regions.

The resulting mask can be expressed as

$$B_{\mathrm{morph}} = (B \circ K) \bullet K, \tag{9}$$

where $\circ$ represents morphological opening and $\bullet$ represents morphological closing.

3.6 Largest Connected Component Selection

The processed binary mask may still contain multiple disconnected foreground regions. Since the experimental setting assumes a single dominant skin lesion in each image, connected-component analysis is used to retain the largest foreground component.

Let the connected foreground components be

$$C_1, C_2, \ldots, C_n. \tag{10}$$

The component with the largest number of pixels is selected:

$$C_{\text{largest}} = \arg\max_{C_i}; |C_i|. \tag{11}$$

The final predicted segmentation mask is therefore

$$P = C_{\text{largest}}. \tag{12}$$

This assumption simplifies the classical baseline but also represents one of its limitations: unrelated dark structures that form a larger connected region than the lesion may be incorrectly selected.

3.7 Evaluation Procedure

The classical segmentation pipeline is applied independently to all 1,000 images in the held-out test subset.

For each test image, the predicted binary mask is compared with the corresponding ground-truth mask using IoU and Dice.

The final performance is obtained by averaging the per-image metrics across the complete test set.

3.8 Results

The classical Otsu baseline obtained the results shown in Table 3.

Table 3: Performance of the classical Otsu baseline on the 1,000-image test set.

Metric	**Mean**	**Standard deviation**
IoU	0.486586	0.351457
Dice	0.565993	0.378914

The median IoU was 0.599406, with values ranging from 0.000000 to 0.967922. The median Dice coefficient was 0.749535, with values ranging from 0.000000 to 0.983700.

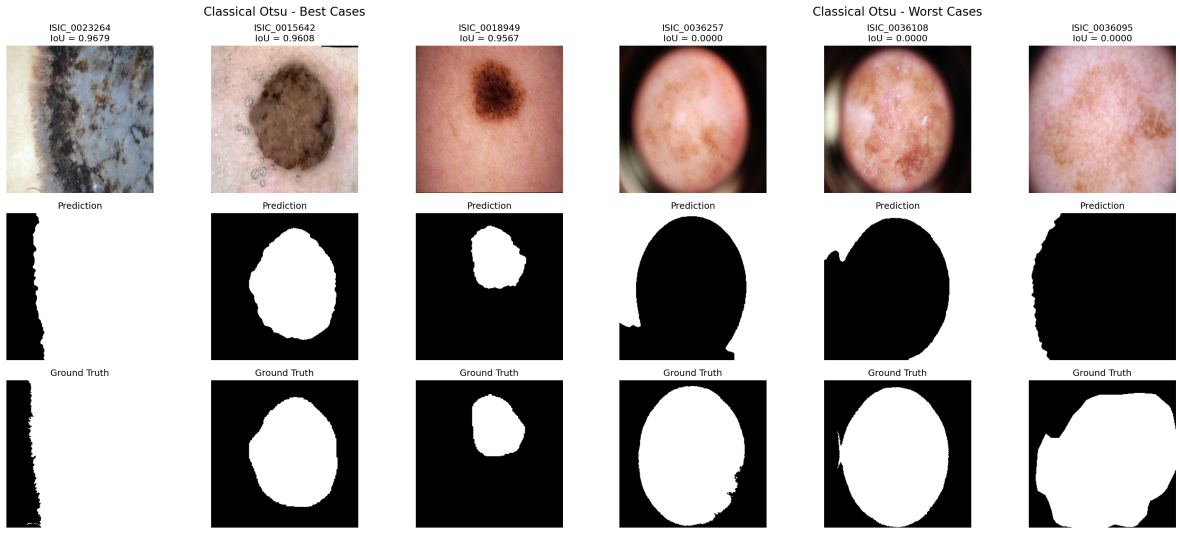
The relatively large standard deviations indicate substantial variation in performance between individual images. Some images can be segmented relatively well using intensity thresholding, while others are poorly segmented or completely missed.

Table 4: Additional statistics for the classical Otsu baseline.

Metric	Median	Minimum	Maximum
IoU	0.599406	0.000000	0.967922
Dice	0.749535	0.000000	0.983700

3.9 Qualitative Analysis

Figure 3 presents representative examples of the classical segmentation results.



(a) Best segmentation (b) Worst segmentation

Figure 1: Representative examples of classical Otsu segmentation.

The qualitative results illustrate several typical behaviors of the classical method. When the lesion has a sufficiently distinct intensity from the surrounding skin, thresholding can produce a reasonable approximation of the lesion boundary. However, images containing weak contrast, illumination variations, hair, or other dark structures can lead to substantial segmentation errors.

These limitations are expected because the method relies primarily on pixel intensity rather than learning the visual characteristics and spatial structure of skin lesions.

3.10 Discussion

The Otsu baseline provides an important reference for the subsequent experiments. Its mean IoU of 0.486586 and mean Dice of 0.565993 demonstrate that a simple intensity-based approach can provide a non-trivial segmentation estimate, but its performance varies considerably across images.

The baseline also reveals an important limitation of purely automatic threshold-based segmentation: the same general procedure must operate across lesions with different colors, shapes, contrasts, and surrounding skin appearances.

This limitation motivates the use of learned segmentation models in the subsequent experiments. In particular, the next experiment investigates whether the pretrained Segment Anything Model (SAM), despite not being trained specifically for dermoscopic images, can provide improved segmentation when supplied with an appropriate spatial prompt.

3.11 Summary

The main findings of Step 1 are:

- The classical Otsu pipeline provides a simple, fully automatic segmentation baseline.
- The method achieves a mean IoU of 0.486586 and a mean Dice coefficient of 0.565993 on the held-out test set.
- Performance varies substantially between individual images.
- Hair, low lesion-to-background contrast, and other image variations can negatively affect the segmentation.
- The baseline does not use any learned representation or prior knowledge about skin lesions.

These results establish the reference performance against which the SAM-based approaches introduced in Step 2 are evaluated.

4 Step 2: Zero-Shot Segment Anything Model

The second experiment evaluates the pretrained Segment Anything Model (SAM) without task-specific fine-tuning.

The SAM ViT-B model is used with its pretrained weights. The image encoder, prompt encoder, and mask decoder are all kept at their pretrained values.

Two prompting strategies are evaluated.

4.1 Ground-Truth Bounding Box

In the first configuration, the bounding box surrounding the ground-truth lesion mask is used as the SAM prompt. This experiment is deliberately not considered a deployable segmentation pipeline because the bounding box is computed directly from the ground-truth annotation. Instead, it measures how well pretrained SAM can segment the lesion when provided with an accurate spatial localization of the target.

This configuration represented as:

$$\text{Image} \rightarrow \text{GT mask} \rightarrow \text{GT bounding box} + \text{pre-trained SAM} \rightarrow \text{final mask}.$$

The resulting mean IoU was 0.775394, with a mean Dice coefficient of 0.865089.

4.2 Automatically Generated Otsu Bounding Box

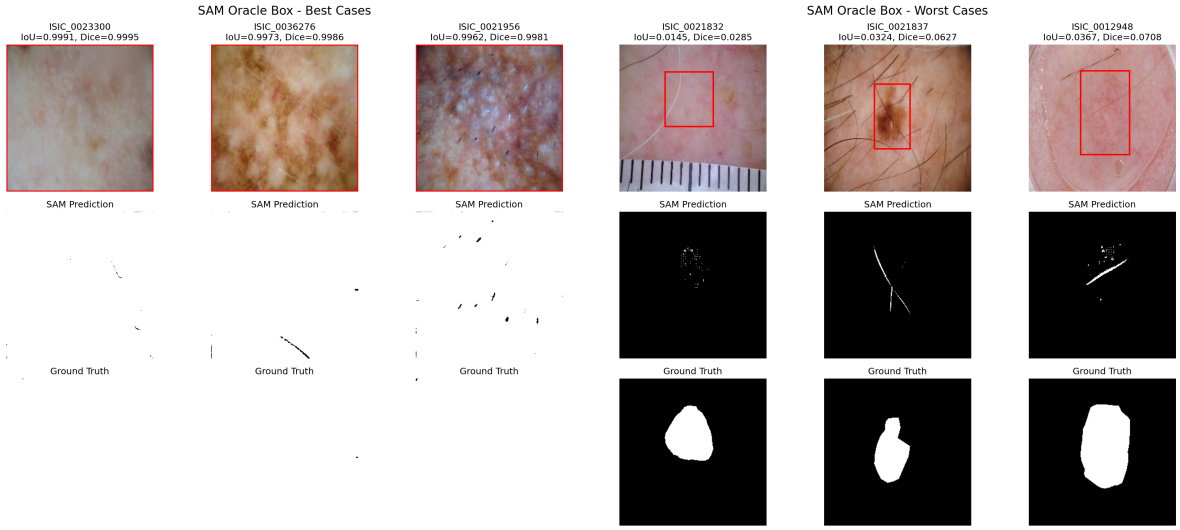
The second configuration removes access to the ground-truth mask at inference time.

The Otsu baseline produces a binary segmentation, from which the bounding box surrounding the detected foreground region is computed. This bounding box is then provided to SAM as its prompt.

This configuration represents a fully automatic pipeline:

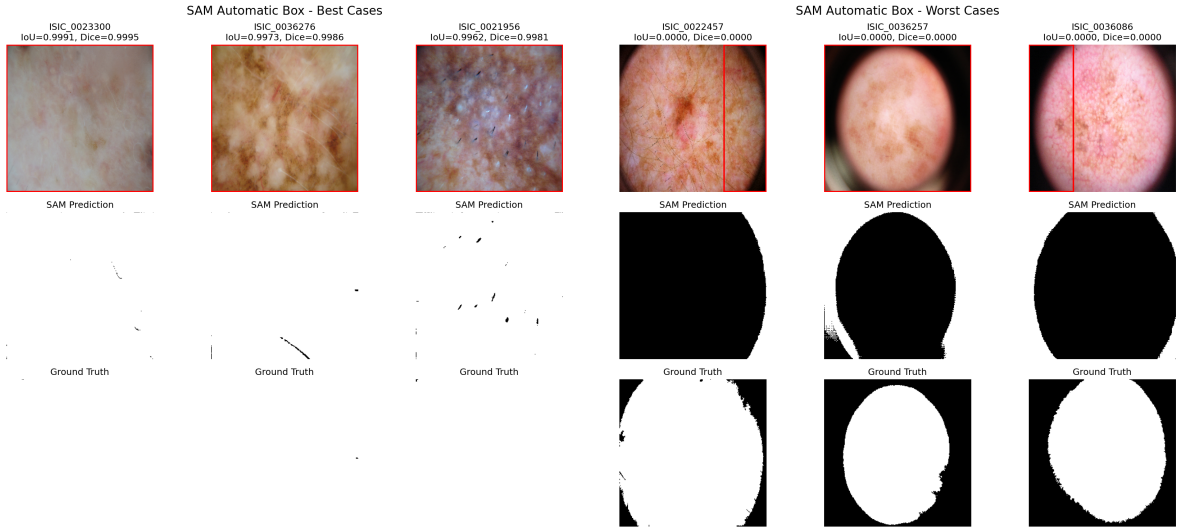
$$\text{Image} \rightarrow \text{Otsu segmentation} \rightarrow \text{Otsu bounding box} + \text{pre-trained SAM} \rightarrow \text{final mask}.$$

The resulting mean IoU was 0.479979, while the mean Dice coefficient was 0.561653.



(a) Best segmentation (b) Worst segmentation

Figure 2: Representative examples of SAM with oracle bounding box.



(a) Best segmentation (b) Worst segmentation

Figure 3: Representative examples of SAM with classical Otsu bounding box.

The substantial difference between the ground-truth-box and Otsu-box configurations demonstrates that the quality of the prompt is an important factor in SAM's segmentation performance for this task.

5 Step 3: Fine-Tuning the SAM Mask Decoder

The third stage investigates whether SAM can be adapted to the ISIC domain through task-specific training.

Instead of fine-tuning the entire SAM architecture, only the mask decoder is optimized, while both the image encoder and the prompt encoder remain frozen.

This design has two advantages. First, it considerably reduces the number of trainable param-

eters compared with full-model fine-tuning. Second, it allows the experiment to specifically investigate whether the pretrained image representation and prompt representation can be retained while adapting the final segmentation component to dermoscopic images.

5.1 Step 3A: Fine-Tuning with Ground-Truth Boxes

The first fine-tuning experiment uses ground-truth lesion bounding boxes as prompts during training.

The training procedure can be summarized as

$$\begin{array}{c} \text{Image} \\ \downarrow \\ \text{frozen image encoder} \\ \downarrow \\ \text{frozen prompt encoder} \\ \downarrow \\ \text{trainable mask decoder} \\ \downarrow \\ \text{predicted mask} \end{array}$$

The 2,594 training images are processed using a data loader. The 100-image validation set is used to select the best-performing checkpoint. The 1,000-image test set remains untouched until final evaluation.

This configuration represented as:

$$\text{Image} \rightarrow \text{GT mask} \rightarrow \text{GT bounding box} + \text{fine-tuned SAM} \rightarrow \text{final mask}.$$

After fine-tuning, the model achieved a mean IoU of 0.891804 and a mean Dice coefficient of 0.941273 when evaluated with ground-truth bounding-box prompts.

Table 5: Comparison of segmentation performance between pretrained and fine-tuned SAM using ground-truth bounding boxes.

Method	**Mean IoU**	**Std IoU**	**Mean Dice**	**Std Dice**
Pretrained SAM + GT box	0.7780	0.1614	0.8631	0.1351
Fine-tuned SAM + GT box	0.8918	0.0680	0.9413	0.0431

For comparison, the corresponding pretrained SAM configuration achieved a mean IoU of 0.7784. Thus, task-specific training of the mask decoder substantially improved segmentation performance under the same ground-truth-box prompting condition.

5.2 Step 3B: Fine-Tuned SAM with Automatically Generated Boxes

The second fine-tuning experiment evaluates whether the adapted SAM model can benefit from the automatically generated Otsu bounding boxes.

At inference time, the pipeline is

$$\text{Image} \rightarrow \text{Otsu segmentation} \rightarrow \text{Otsu bounding box} + \text{fine-tuned SAM} \rightarrow \text{final mask}.$$

The image encoder and prompt encoder remain frozen, while the mask decoder uses the weights obtained during Step 3A.

Table 6: Comparison of segmentation performance across different methods.

Method	Mean IoU	Mean Dice
Classical Otsu	0.4866	0.5660
Pretrained SAM + GT box	0.7754	0.8651
Pretrained SAM + automatic Otsu box	0.4800	0.5617
Fine-tuned SAM + GT box	0.8918	0.9413
Fine-tuned SAM + automatic Otsu box	0.5490	0.6249

This configuration achieved a mean IoU of 0.549019 and a mean Dice coefficient of 0.624872.

Compared with the classical Otsu baseline, whose mean IoU was 0.486586, the fine-tuned SAM pipeline improved the mean IoU by 0.062433. Compared with pretrained SAM using the same Otsu-derived prompt, the improvement was 0.069040 in mean IoU.

The results therefore separate two effects: adapting SAM to the ISIC domain improves the decoder's behavior, but the automatically generated prompt remains a significant source of error.

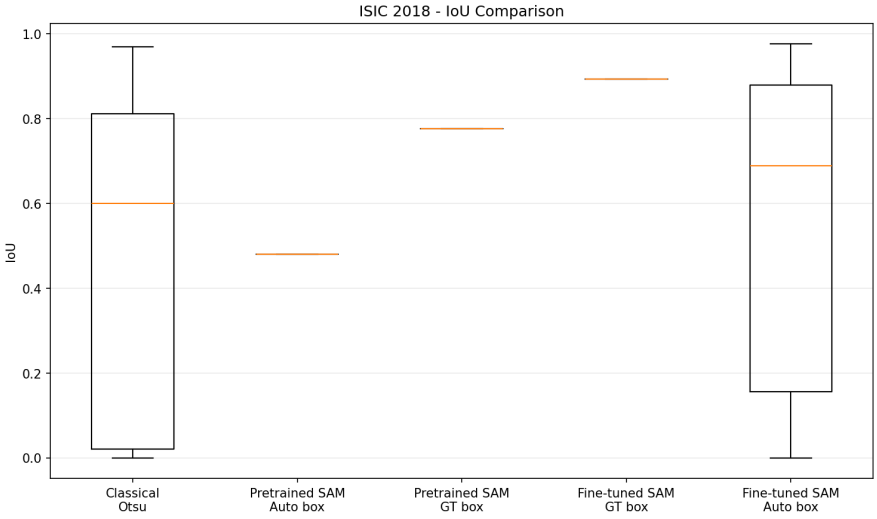


Figure 4: IoU comparison from steps 1, 2 and 3.

6 Step 4: Interactive Segmentation

The fourth experiment investigates a human-in-the-loop segmentation strategy.

The motivation is that the Otsu segmentation can provide a useful initial estimate but may contain incorrect regions or miss parts of the lesion. Instead of requiring a user to manually annotate every image, the proposed procedure only requests interaction when the automatically generated segmentation differs sufficiently from the ground-truth segmentation during experimental analysis.

The conceptual pipeline is

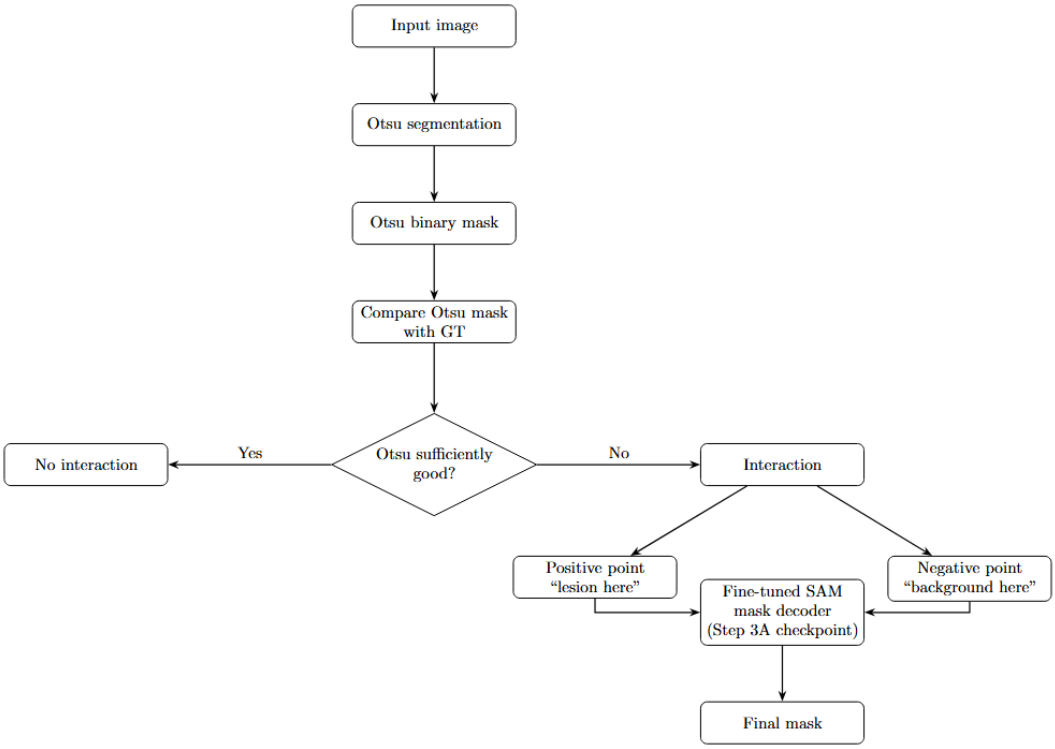


Figure 5: Conceptual pipeline of Interactive Segmentation.

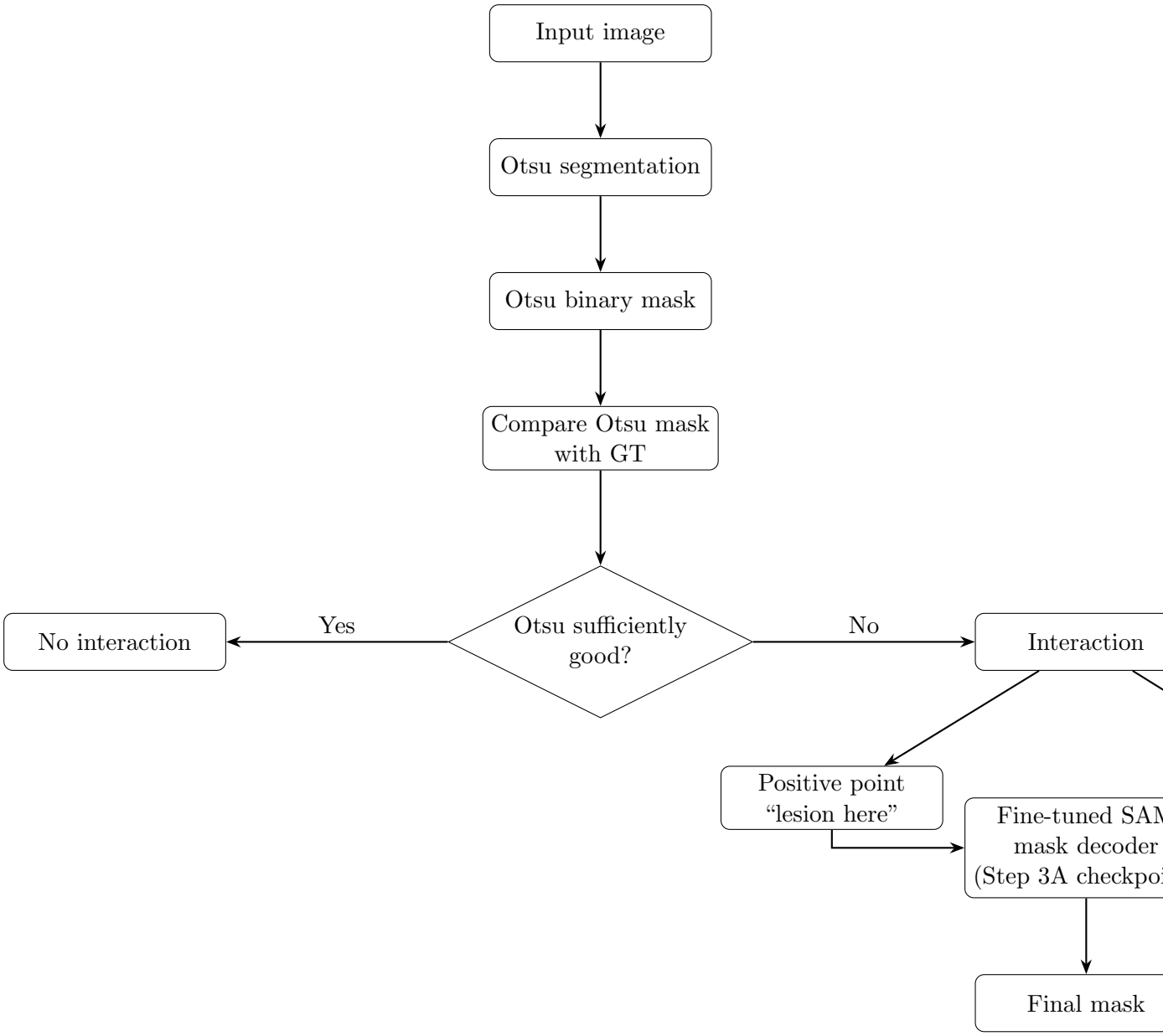


Figure 6: Overview of the proposed segmentation pipeline. The Otsu segmentation is first compared with the ground-truth mask. If the segmentation is sufficiently accurate, no user interaction is required. Otherwise, positive and negative points are provided to the fine-tuned SAM mask decoder.

For the interactive correction, at least one positive point and one negative point are used. The positive point indicates a location that belongs to the lesion, while the negative point indicates a location that should not be considered part of the lesion. These points are provided to SAM together with the image, allowing the model to refine the segmentation according to the user's correction.

The purpose of this experiment is primarily to investigate the interaction mechanism rather than to establish a final benchmark score. In particular, it explores whether a very small amount of human input can correct an imperfect automatic segmentation and provide useful prompts to the fine-tuned SAM decoder.

It is important to distinguish this experimental procedure from a fully automatic evaluation: the use of the ground-truth mask to determine whether interaction is required is appropriate for studying the mechanism experimentally, but would not be available in a real deployment scenario.



(a) Image segmented before human intervention

(b) Interactive SAM with positive and negative points

(c) Image segmented after interactive segmentation

Figure 7: Steps of Interactive Segmentation.

7 Step 5: U-Net Comparison Model

The final experiment uses a conventional supervised segmentation architecture as a comparison point. A U-Net model is trained on the same 2,594-image training set and evaluated using the same 100-image validation set and 1,000-image test set. The U-Net uses a ResNet-34 encoder initialized with ImageNet-pretrained weights. The decoder is trained for the ISIC lesion segmentation task.



Figure 8: U-Net training history

The model receives a resized image as input and directly predicts a binary lesion mask. Unlike SAM, no bounding-box or point prompt is required during inference. The training objective combines binary cross-entropy and Dice loss. The best model checkpoint is selected according to validation IoU.

The resulting model achieved a mean IoU of 0.810632 and a mean Dice coefficient of 0.886213 on the test set.

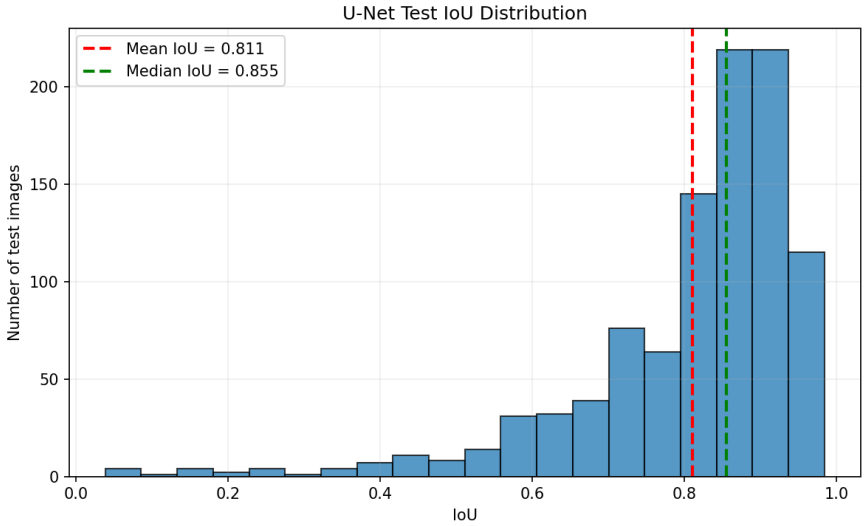


Figure 9: U-Net IoU distribution.

This experiment provides a useful architectural comparison because it represents a conventional supervised segmentation approach trained specifically for the target task, rather than a general-purpose segmentation foundation model.

8 Results

8.1 Overall Quantitative Results

The five experimental stages provide a progressive comparison between classical image processing, a pretrained foundation model, task-specific adaptation of the foundation model, interactive segmentation, and a conventional supervised segmentation architecture.

The quantitative results obtained on the 1,000-image test set are summarized in Table 7.

Table 7: Comparison of the evaluated segmentation approaches on the ISIC 2018 Task 1 test set.

Method	**Mean IoU**	**Mean Dice**
Classical Otsu	0.486586	0.565993
Pretrained SAM + GT box	0.775394	0.865089
Pretrained SAM + automatic Otsu box	0.479979	0.561653
Fine-tuned SAM + GT box	0.891804	0.941273
Fine-tuned SAM + automatic Otsu box	0.549019	0.624872
U-Net + ResNet-34 ImageNet encoder	0.810632	0.886213

The results reveal an important distinction between the quality of the segmentation model and the quality of the prompt supplied to SAM. In particular, SAM performs substantially better when supplied with an accurate ground-truth bounding box than when its bounding box is generated automatically from the Otsu segmentation.

8.2 Classical Otsu Baseline

The classical Otsu method obtained a mean IoU of 0.486586 and a mean Dice coefficient of 0.565993.

This result establishes the reference level for the subsequent experiments. The relatively large standard deviations observed for the baseline indicate that its performance varies considerably between images. In particular, the test-set IoU ranged from 0.000000 to 0.967922.

This variability is expected from a simple threshold-based method because the appearance of dermoscopic images can vary substantially in terms of illumination, lesion contrast, surrounding skin, and image artifacts.

The Otsu baseline nevertheless provides an important component of the subsequent experiments because its predicted mask is also used to generate an automatic bounding box for SAM.

8.3 Pretrained SAM with Ground-Truth Boxes

When pretrained SAM is provided with a ground-truth bounding box, the mean IoU increases to 0.775394, while the mean Dice coefficient reaches 0.865089.

Compared with the classical Otsu baseline, this represents an increase of mean IoU by:

$$0.775394 - 0.486586 = 0.288808.$$

The result demonstrates that the pretrained SAM model contains useful segmentation capabilities that can transfer to the dermoscopic domain without task-specific training, provided that an accurate spatial prompt is available.

However, this experiment should be interpreted as an upper-bound-style experiment rather than as a realistic automatic segmentation pipeline. The bounding box is computed from the ground-truth annotation and therefore provides information that would not normally be available during inference.

8.4 Pretrained SAM with Automatically Generated Boxes

When the SAM bounding box is instead obtained from the Otsu segmentation, the mean IoU decreases substantially to 0.479979, with a mean Dice coefficient of 0.561653.

Interestingly, this result is slightly below the classical Otsu baseline:

$$0.479979 - 0.486586 = -0.006607. \tag{13}$$

The difference indicates that the pretrained SAM model does not automatically improve the segmentation simply because it is given a prompt derived from an existing segmentation method.

The main difference between the two SAM experiments is therefore not the SAM model itself, but the quality of the supplied bounding box. With the ground-truth box, SAM receives an accurate localization of the lesion. With the Otsu-derived box, errors made by the classical segmentation process can be propagated into the SAM prompting stage.

This observation is important for the remainder of the experiments: improving the segmentation model alone may not be sufficient if the automatic prompt remains unreliable.

8.5 Fine-Tuned SAM with Ground-Truth Boxes

Fine-tuning the SAM mask decoder on the ISIC training set produces a substantial improvement when ground-truth bounding boxes are used.

The fine-tuned model achieves:

- mean IoU: 0.891804;
- mean Dice: 0.941273.

The corresponding pretrained SAM configuration achieved a mean IoU of 0.775394. The change is therefore

$$0.891804 - 0.775394 = 0.116410.$$

The measured experiment reported an improvement of approximately 0.114 IoU points depending on the exact evaluation subset and stored comparison values.

The important observation is that adapting only the mask decoder was sufficient to substantially improve the model's performance on the ISIC segmentation task while keeping the image encoder and prompt encoder frozen.

This suggests that the pretrained SAM representation already provides useful information for the task, while the mask decoder benefits from specialization to the visual characteristics of dermoscopic lesions.

8.6 Fine-Tuned SAM with Automatically Generated Boxes

The fine-tuned model was subsequently evaluated using automatically generated Otsu-derived bounding boxes.

The mean IoU increased from 0.479979 with pretrained SAM to 0.549019 after mask-decoder fine-tuning. Similarly, mean Dice increased from 0.561653 to 0.624872.

The corresponding changes are

$$\Delta\text{IoU} = 0.549019 - 0.479979 = 0.069040$$

and

$$\Delta\text{Dice} = 0.624872 - 0.561653 = 0.063219.$$

The fine-tuned model therefore benefits from task-specific adaptation even when the prompt is generated automatically.

However, the performance remains considerably lower than the corresponding ground-truth-box experiment. The difference between the two fine-tuned configurations is

$$0.891804 - 0.549019 = 0.342785$$

in mean IoU.

This gap is one of the most important findings of the study. It indicates that the limitation of the automatic pipeline is not solely the segmentation capability of the SAM decoder. The quality of the automatically generated prompt remains a major factor.

8.7 Interactive Segmentation

The interactive experiment was designed to investigate whether limited human input could address the weaknesses of the automatic Otsu-derived prompt.

The proposed procedure starts from the Otsu segmentation and determines whether the resulting mask differs sufficiently from the ground-truth mask according to a predefined threshold. When interaction is required, the user supplies:

- one positive point indicating a location belonging to the lesion;
- one negative point indicating a location belonging to the background.

These points are then provided to the fine-tuned SAM model as an interactive prompt.

The experiment is primarily intended as a proof-of-concept investigation of the interaction mechanism rather than as another fully automatic benchmark. In particular, the experimental procedure uses the ground-truth mask to determine when interaction is requested. This allows the number of manual interactions to be controlled during experimentation, but it would not be available in a real-world deployment.

The interactive strategy therefore addresses a different question from Steps 1–3: instead of asking whether segmentation can be performed without human intervention, it investigates whether a very small amount of human guidance can correct failures produced by an automatic segmentation pipeline.

8.8 U-Net with ResNet-34 Encoder

The U-Net experiment provides a comparison with a conventional supervised segmentation architecture.

The model uses a ResNet-34 encoder initialized with ImageNet-pretrained weights and is trained directly on the ISIC training set. The model does not require an external bounding-box or point prompt at inference time.

The final test performance was:

- mean IoU: 0.810632;
- mean Dice: 0.886213.

The best checkpoint was obtained at epoch 12, with a validation IoU of 0.824519.

The U-Net therefore provides substantially higher performance than the classical Otsu baseline and both automatically prompted SAM configurations. Its performance is also higher than pretrained SAM with a ground-truth box in terms of mean IoU.

At the same time, its mean IoU remains below the fine-tuned SAM model when the latter is given a ground-truth bounding box.

This comparison illustrates the importance of separating model architecture from prompting conditions. The U-Net performs segmentation directly from the image, whereas the SAM experiments depend on additional spatial prompts. Consequently, the numerical results should be interpreted together with the type of information available to each method.

8.9 Comparison of the Experimental Progression

The progression of the experiments can be summarized as follows:

1. The classical Otsu method establishes a simple non-learning reference.
2. Pretrained SAM demonstrates that a general-purpose segmentation foundation model can perform well on dermoscopic images when supplied with an accurate bounding box.
3. Replacing the ground-truth box with an automatically generated Otsu box causes a substantial reduction in performance, demonstrating the importance of prompt quality.

4. Fine-tuning the SAM mask decoder substantially improves performance with ground-truth prompts.
5. Fine-tuning also improves performance with automatically generated prompts, although the improvement is smaller than in the ground-truth-box setting.
6. The interactive experiment explores whether minimal human input can compensate for imperfect automatic prompts.
7. The U-Net provides a dedicated supervised segmentation reference that does not require prompting during inference.

Overall, the experiments show that the segmentation problem can be viewed as involving two coupled components: *the ability of the model to segment the lesion* and *the ability of the system to provide useful information about where the lesion is located*. The SAM experiments make this distinction particularly visible because the same underlying model produces substantially different results depending on the prompt quality.

9 Discussion

9.1 Main Findings

The experiments provide several complementary observations about skin lesion segmentation on the ISIC 2018 Task 1 dataset.

First, the results demonstrate that a simple classical segmentation method can provide a useful baseline, but its performance is limited by the variability of dermoscopic images. The Otsu method achieved a mean IoU of 0.486586. This establishes that a substantial portion of the segmentation problem cannot be solved reliably using a single global thresholding strategy.

Second, the experiments with pretrained SAM demonstrate the importance of the information provided to a prompt-based segmentation model. When SAM receives a ground-truth bounding box, its mean IoU reaches 0.775394. However, when the bounding box is obtained automatically from the Otsu segmentation, the mean IoU falls to 0.479979.

This difference is important because the SAM model itself is unchanged between these two experiments. The main difference is the quality of the spatial information provided to the model. The experiment therefore indicates that prompt generation is a major component of a practical SAM-based segmentation pipeline.

Third, fine-tuning the SAM mask decoder substantially improves the segmentation performance when accurate bounding boxes are available. The fine-tuned model reaches a mean IoU of 0.891804 with ground-truth boxes, compared with 0.775394 for pretrained SAM under the same prompting condition.

This suggests that the general-purpose SAM representation can be adapted effectively to the characteristics of dermoscopic images without fine-tuning the complete model. In the present experiments, the image encoder and prompt encoder were kept frozen while only the mask decoder was optimized.

9.2 The Importance of Prompt Quality

One of the clearest observations from the experiments is that model quality and prompt quality cannot be considered independently.

For pretrained SAM, replacing the ground-truth box with an Otsu-derived box changes the mean IoU from 0.775394 to 0.479979. A similar phenomenon remains after fine-tuning: the fine-tuned model achieves 0.891804 with ground-truth boxes but 0.549019 with automatically generated boxes.

The difference between these configurations suggests that the automatic pipeline has two distinct sources of error:

1. errors in estimating the lesion location before SAM is invoked;
2. errors in the subsequent SAM segmentation.

The first source of error is particularly important because an inaccurate bounding box can limit the information available to the segmentation model before the mask decoder produces its prediction.

This also explains why fine-tuning alone does not completely solve the automatic segmentation problem. Fine-tuning improves the behavior of the mask decoder, but it does not eliminate errors introduced by the Otsu-based prompt-generation stage.

9.3 Effect of Fine-Tuning

The comparison between pretrained and fine-tuned SAM provides evidence that domain-specific adaptation is beneficial for this task.

With ground-truth bounding boxes, fine-tuning increases the mean IoU from 0.775394 to 0.891804. With automatically generated Otsu boxes, the mean IoU increases from 0.479979 to 0.549019.

The improvement is therefore observed under both prompting conditions, although the absolute performance remains much higher when the bounding box is accurate.

An important aspect of this experiment is that only the mask decoder was trained. The image encoder and prompt encoder remained frozen. This reduces the scope of adaptation and makes the experiment useful for investigating whether the pretrained SAM representation can be retained while adapting the final segmentation component to a specific medical-imaging domain.

However, these results should not be interpreted as demonstrating that mask-decoder-only fine-tuning is universally sufficient. The conclusion is specific to the present dataset, model configuration, training procedure, and prompting strategy.

9.4 Comparison with the U-Net Architecture

The U-Net experiment provides a different perspective on the problem.

The U-Net with a ResNet-34 ImageNet-pretrained encoder achieved a mean IoU of 0.810632. Unlike SAM, the U-Net directly predicts a segmentation mask from the input image and does not require a bounding-box or point prompt.

This makes the U-Net an important comparison because it represents a conventional supervised segmentation approach rather than a prompt-based foundation model.

The U-Net result also illustrates that a model specifically trained for the target segmentation task can achieve strong performance without requiring an external prompt. At the same time, the fine-tuned SAM configuration with ground-truth boxes achieved a higher mean IoU of 0.891804.

These two experiments should therefore not be interpreted as a simple architecture-versus-architecture comparison. The U-Net and SAM experiments receive different types of information at inference time. The U-Net receives only the image, whereas the SAM experiments additionally receive spatial prompts.

Consequently, the comparison is useful for understanding different segmentation paradigms, but it does not constitute a controlled comparison in which all input information is identical.

9.5 Interactive Segmentation as a Middle Ground

The interactive segmentation experiment addresses a limitation observed in the automatically prompted SAM experiments.

The fully automatic pipeline has the advantage of requiring no human input, but the Otsu-derived bounding box can be inaccurate. Conversely, using the ground-truth bounding box produces much stronger SAM performance but requires information that is unavailable during real inference.

Interactive segmentation provides a possible middle ground between these two situations.

Instead of requiring the user to draw an entire segmentation mask or bounding box, the proposed experiment uses only two points:

- one positive point indicating lesion tissue;
- one negative point indicating background tissue.

The objective is therefore to investigate whether a small amount of human information can compensate for an imperfect automatic prompt.

This approach is particularly relevant for applications in which complete automation is not mandatory but reducing the amount of manual annotation is desirable. A clinician or annotator could potentially provide a very small number of corrections instead of manually outlining the entire lesion.

Nevertheless, the current experimental implementation should be regarded as a proof-of-concept. The decision to request human interaction is made using the ground-truth mask during the experiment. This is appropriate for controlled experimentation but would introduce information leakage if used as part of a real deployment evaluation.

A future version of the experiment should therefore investigate interaction-triggering criteria that do not require access to the ground-truth annotation.

9.6 Relationship Between the Experiments

The experiments can be viewed as a progression from fully automatic but simple segmentation toward increasingly informed segmentation systems.

The classical Otsu approach uses only image intensity information. The automatic SAM pipeline adds a foundation model but still depends on an automatically generated spatial prompt. Fine-tuning adds domain-specific training to the foundation model. Interactive segmentation then introduces a small amount of human information to correct the automatically generated prompt.

The resulting conceptual progression is:

Classical segmentation
↓
Automatic prompt
↓
Foundation model
↓
Domain adaptation
↓
Human correction

The experiments consequently show that segmentation quality is not determined by a single component. It depends on the interaction between the image representation, the segmentation model, the training procedure, and the information supplied through the prompt.

9.7 Limitations

Several limitations should be considered when interpreting the results.

9.7.1 Limited Validation Set

The validation set contains only 100 images, compared with 2,594 training images and 1,000 test images. Although this validation set was kept separate from training, its relatively small size may make model-selection measurements more variable.

9.7.2 Fixed Dataset Split

All experiments use the same train, validation, and test split. This makes the experiments directly comparable within this project, but it does not measure the variability that might result from using different dataset partitions.

9.7.3 Prompt Dependence of SAM

The SAM experiments depend strongly on the quality of the prompt. The ground-truth-box experiments provide useful information about the segmentation capability of SAM, but they do not represent a fully automatic application scenario.

Similarly, the automatically generated Otsu box represents only one possible prompt-generation strategy. Other automatic localization methods could produce different results.

9.7.4 Interactive Experiment Design

The interactive experiment uses the ground-truth annotation to determine whether a correction is necessary. This makes it unsuitable for claiming fully autonomous operation. Its purpose in the current study is to investigate the feasibility of minimal human interaction.

A more realistic future experiment would define a model-based or uncertainty-based criterion for requesting user intervention without using the ground-truth mask.

9.7.5 Single SAM Model Configuration

The experiments use SAM ViT-B. Other SAM model sizes or newer segmentation foundation models could behave differently. The conclusions should therefore be understood as applying to the tested model configuration rather than to all foundation models.

9.7.6 U-Net Configuration

The U-Net experiment uses a ResNet-34 encoder initialized with ImageNet-pretrained weights. Other encoder architectures, augmentation strategies, loss functions, input resolutions, and training schedules could produce different results.

Therefore, the U-Net experiment should be interpreted as a representative dedicated segmentation baseline rather than an exhaustive optimization of U-Net architectures.

9.8 Reproducibility Considerations

The experiments were designed to maintain a common evaluation protocol across methods. In particular, the same 1,000-image test set was used for the final quantitative evaluation of the different approaches.

For the learning-based experiments, the training and validation data were separated from the test data. Model selection was performed using the validation set, while the test set was reserved for final evaluation.

The experiments also saved per-image predictions and quantitative results, allowing the aggregate metrics to be reproduced and individual failure cases to be inspected.

This is particularly important for segmentation because mean IoU and mean Dice alone do not describe how a model behaves on individual images. Qualitative inspection of successful and unsuccessful cases can reveal failure modes that are hidden by aggregate statistics.

10 Conclusion and Future Work

10.1 Conclusion

This work investigated several approaches to skin lesion segmentation on the ISIC 2018 Task 1 dataset, progressing from a classical image-processing baseline to foundation-model-based segmentation, task-specific fine-tuning, interactive segmentation, and a conventional supervised segmentation architecture.

The classical Otsu method provided a simple reference point, achieving a mean IoU of 0.486586. Although computationally simple and fully automatic, its performance varied substantially across images, illustrating the difficulty of separating lesions from surrounding skin using intensity thresholding alone.

The experiments with the pretrained Segment Anything Model demonstrated the potential of a general-purpose segmentation foundation model for this task. When supplied with a ground-truth bounding box, pretrained SAM achieved a mean IoU of 0.775394. However, replacing the ground-truth box with an automatically generated Otsu-derived box reduced the mean IoU to 0.479979. This experiment highlighted the importance of prompt quality in prompt-based segmentation.

Fine-tuning the SAM mask decoder provided a substantial improvement when accurate spatial prompts were available. With ground-truth bounding boxes, the fine-tuned model achieved a mean IoU of 0.891804 and a mean Dice coefficient of 0.941273. Importantly, this adaptation was performed while keeping the image encoder and prompt encoder frozen.

The automatically prompted fine-tuned SAM configuration achieved a mean IoU of 0.549019. Although this was an improvement over both the pretrained SAM with an Otsu-derived box and the classical Otsu baseline, the performance remained substantially below that obtained

with ground-truth boxes. This reinforces the observation that prompt generation represents a major bottleneck in the automatic SAM pipeline.

The interactive segmentation experiment was introduced to investigate a possible solution to this bottleneck. Rather than requiring a complete manual segmentation, the proposed approach uses a minimal interaction consisting of one positive and one negative point. The purpose of this experiment was to investigate whether such limited human input can provide useful corrective information to the fine-tuned SAM model.

Finally, the U-Net experiment provided a dedicated supervised segmentation reference. The U-Net with an ImageNet-pretrained ResNet-34 encoder achieved a mean IoU of 0.810632 and a mean Dice coefficient of 0.886213 on the same test set.

Taken together, the experiments demonstrate that skin lesion segmentation can be approached from several complementary perspectives. The results also show that the quality of the information provided to a segmentation model can be as important as the segmentation model itself. In particular, the large difference between ground-truth and automatically generated SAM prompts demonstrates that improving the localization or prompting stage is an important direction for future development.

10.2 Future Work

Several extensions could build upon the experiments presented in this work.

10.2.1 Improved Automatic Prompt Generation

The largest limitation identified in the SAM experiments is the quality of the automatically generated prompt. The current pipeline relies on Otsu thresholding to generate the bounding box.

Future work could investigate more robust automatic localization techniques, such as:

- classical image-processing methods specifically designed for dermoscopic images;
- lightweight object-detection models;
- a dedicated lesion-localization network;
- a small segmentation network used only to generate SAM prompts;
- combinations of color, texture, and morphological information.

The objective would be to generate a more reliable prompt before invoking SAM.

10.2.2 More Realistic Interactive Segmentation

The interactive experiment could be extended into a more realistic human-in-the-loop system.

The current experimental design uses the ground-truth mask to determine whether human intervention is necessary. In a real application, this information would not be available.

A future system could instead estimate segmentation uncertainty or detect potentially unreliable predictions automatically. User interaction could then be requested only when the system identifies a low-confidence or ambiguous segmentation.

The number and type of interactions could also be systematically studied. For example, experiments could compare:

- one positive point;

- one negative point;
- one positive and one negative point;
- multiple corrective points;
- iterative point-based correction.

Such experiments would make it possible to study the relationship between the amount of human effort and the resulting segmentation quality.

10.2.3 Training with Interactive Prompts

The current SAM fine-tuning experiment uses ground-truth bounding boxes for training. A natural extension would be to train the mask decoder using simulated interactive prompts.

For example, positive and negative points could be generated automatically from the ground-truth mask during training. Different point-generation strategies could simulate realistic user corrections.

The model could then learn to respond specifically to point-based corrective interactions rather than relying primarily on bounding-box prompts.

10.2.4 Joint Optimization of Prompt Generation and Segmentation

The experiments suggest that the current pipeline contains two separate components:

$$\text{automatic localization} \rightarrow \text{SAM segmentation}. \tag{14}$$

Future work could investigate whether these components can be optimized jointly.

A dedicated localization network could first estimate the lesion region and generate a prompt, while SAM performs the final segmentation. The localization component could potentially be trained using bounding-box annotations derived from the available segmentation masks.

This would provide a more principled alternative to the current Otsu-based prompt-generation stage.

10.2.5 Comparison with Additional Segmentation Architectures

The U-Net experiment provides one dedicated-architecture comparison point. Future experiments could extend this comparison to additional architectures such as U-Net variants, DeepLab-based models, transformer-based segmentation models, and other modern medical-image segmentation architectures.

Such comparisons would help determine whether the observed differences are specific to U-Net or reflect broader differences between prompt-based foundation models and task-specific segmentation networks.

10.2.6 Additional Evaluation Metrics

IoU and Dice provide useful measures of segmentation overlap, but additional metrics could provide a more complete analysis.

Future work could include:

- precision and recall;

- sensitivity and specificity;
- Hausdorff distance;
- boundary-based metrics;
- lesion-area error;
- inference time;
- GPU memory consumption.

These measurements would be particularly useful for evaluating interactive systems, where segmentation quality should be considered together with computational cost and human interaction time.

10.2.7 Evaluation Across Multiple Data Splits

The current experiments use a fixed train/validation/test split. This ensures that all methods are evaluated under the same conditions, which is useful for direct comparison within this study.

However, evaluating the methods across multiple independent splits or using cross-validation would provide information about the stability of the observed results.

This would be especially useful given the relatively small validation set used in the current experiments.

10.3 Final Perspective

The main contribution of this experimental study is not simply the identification of a single segmentation score. Rather, the experiments provide a progressive investigation of how different sources of information affect skin lesion segmentation.

The experimental progression can be summarized as:

$$\boxed{\text{Classical segmentation} \rightarrow \text{Foundation model} \rightarrow \text{Domain adaptation} \rightarrow \text{Interactive correction}} \tag{15}$$

The results indicate that pretrained segmentation models can provide strong representations for medical-image segmentation, but their practical performance depends strongly on how the target region is localized and communicated to the model.

The study therefore motivates a hybrid perspective in which classical image processing, learned segmentation models, foundation models, and limited human interaction are not necessarily competing alternatives. Instead, they can be combined into a pipeline in which each component addresses a different part of the segmentation problem.

In particular, the results motivate further investigation of systems that combine automatic lesion localization with SAM-based segmentation and optional minimal human correction. Such a system could potentially retain the efficiency of automatic segmentation while providing a mechanism for correcting difficult cases without requiring complete manual annotation.